

## Multiple Choice (10 marks)

1. Which muscle is the most active during a right lateral excursion of the mandible?
  - (a) Left masseter muscle
  - (b) Right buccinator muscle
  - (c) Right temporalis muscle
  - (d) Right masseter muscle
  - (e) Right lateral pterygoid muscle
2. If the frequency of males affected with an X-linked recessive condition in a human population is .10 (one in ten), what will be the expected frequency of affected females?
  - (a) 0.001
  - (b) 0.01
  - (c) 0.1
  - (d) 0.15
  - (e) 0.0011
3. The proportion of babies that have an abnormality identifiable at birth is \_\_\_\_\_.
  - (a) 1 in 100
  - (b) 1 in 50
  - (c) 1 in 5
  - (d) 1 in 10
  - (e) 1 in 1000
4. How are antibody producing cells stimulated?
  - (a) Interacting with a virus short peptide
  - (b) Interacting with a T helper cell.
  - (c) Interacting with a plasma cell
  - (d) Interacting with a virus infected cell
5. Which of the following is most strongly associated with loss of bone density?
  - (a) Environmental factors such as exposure to radiation
  - (b) Genetic factors
  - (c) Psychological factors such as stress levels
  - (d) Atmospheric factors such as air pollution
  - (e) Socioeconomic factors such as income level

## True / False (10 marks)

*Answer True or False*

6. True or False: Enlargement of the submandibular lymph nodes often indicates potential infection sites in the hard palate.
7. True or False: The ureter is the primary structure responsible for collecting urine in the body before it reaches the bladder.
8. True or False: The risk of abnormality in the child of a mother with untreated phenylketonuria is approximately 25%.
9. True or False: The elderly often play a crucial role in facilitating effective change within a community due to their experience and influence.
10. True or False: Research has shown that prenatal maternal stress is a significant factor in social epigenetics, influencing fetal development.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the role of ectomesenchyme in the development of melanocytes, including their origin, function, and implications for health and disease.
12. Discuss the role of the low-density lipoprotein receptor in the pathophysiology of familial hypercholesterolaemia and analyze how mutations in this gene impact cholesterol metabolism.

*End of Paper*